

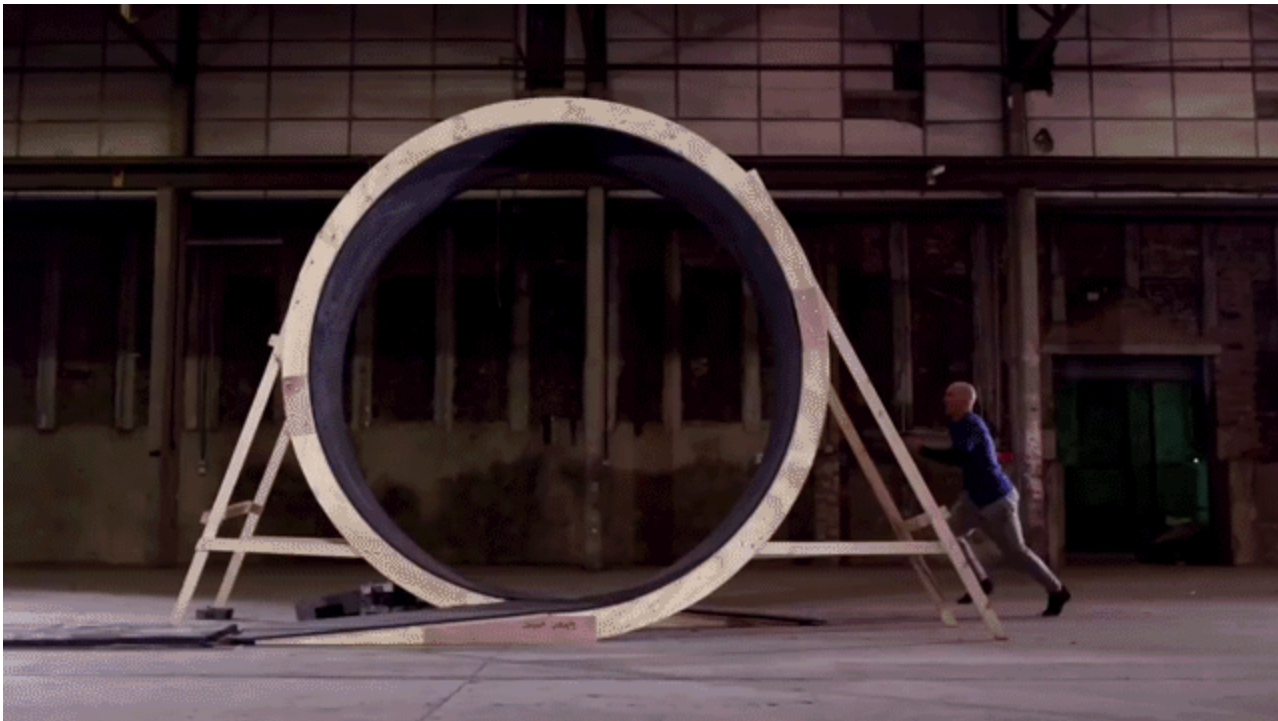
mini assignment conservation energy - loopy loop again



⚠ This is a preview of the published version of the quiz

Started: May 7 at 2:36pm

Quiz Instructions

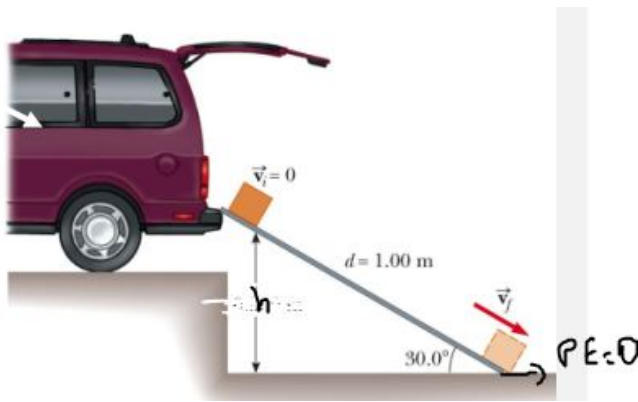


Question 1 20 pts

- 5
- ☐ A 5-kg crate slides down a ramp. The ramp is 1 m in length and inclined at an angle of 30° as shown. The crate starts from rest at the top. The surface friction can be negligible. Use energy methods to determine the speed of the crate at the bottom of the ramp.

hint: $PE_1 + KE_1 = PE_2 + KE_2$ take $g = 10 \text{ m/s}^2$. Also use trig to find $h = d \times \sin(30)$

take $PE = 0$ on the floor.


☐

50m/s

☐

10m/s

☐

3.2 m/s

☐

7m/s

☐

Question 2 20 pts

5
☐ A ~~5~~-kg crate slides down a ramp. The ramp is 1 m in length and inclined at an angle of 30° as shown. The crate starts from rest at the top. The surface in contact have a coefficient of kinetic friction of ~~0.15~~. Use energy methods to determine the speed of the crate at the bottom of the ramp. 0.2

hints : (try without) . use $g = 10\text{m/s}^2$

Mechanical energy 1 - work done by friction = Mechanical energy 2

$PE_1 + KE_1 - (\text{friction} \times \text{distance } 1\text{m}) = PE_2 + KE_2$

friction = $0.2 \times \text{normal}$ by definition and in magnitude

You have memorized that $m g \cos(\text{angle})$ is the perpendicular component of the weight. so what is the normal force in this case? up = down

☐

about 6.5m/s



about 3.5 m/s



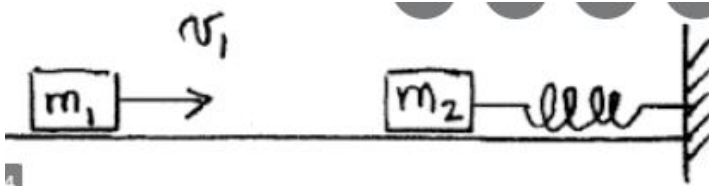
about 2.5m/s



about 4.5m/s



Question 3 20 pts



block m_1 (2kg) has a speed of 2m/s @ right. It collides with m_2 . There is no friction. The spring constant is 50N/m what is the maximum compression x_c in the spring?

hints: Let's consider the system m_1 and m_2 (and Earth). before we only have KE from m_1 . After we only have elastic PE ($0.5 K X^2$). The system m_1+m_2 is at rest.



about 40cm



about 10cm



about 20cm



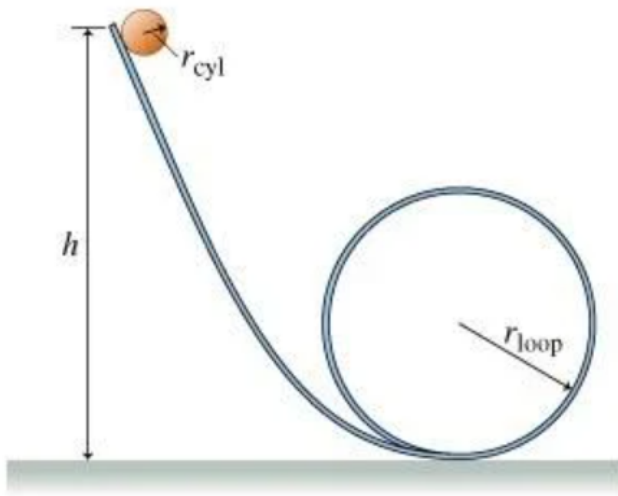
about 5cm



Question 4 20 pts

Do you remember the loopy loop? You used the conservation of mechanical energy and the definition of the centripetal force to find the minimum h so the block makes it around the loop. Same question but for a spinning ball.

What if, instead of a block, we have a marble. The first equation needs to be modified because its rolling. If its rolling it has rotational kinetic energy (spinning energy) in addition to linear KE (its spinning and it is moving forward).



So before: $E_{\text{total}} = mgh$

At the top of the loop after : $E_{\text{total}} = mg(2R) + 0.5mV^2 + 0.5 (2/5) m V^2$ Eq1

note you can simplify with $0.5 (2/5) m V^2 = (1/5) m V^2$

Also at the top. for minimum speed $N=0$ so $mg = mV^2/R$ (centripetal force) Eq2

from Eq2 solve for V^2 and plug V^2 in Eq1.

Then use the conservation of energy to solve for h

What is h ?

☐

$3/10 R$

☐

$27/10 R$

☐

I don't remember how to add fractions

☐

$2R$

☐

$5/2 R$

☐

Question 5 10 pts

When a planet is in orbit around the sun (or a moon around a planet).

PEg is the potential energy of the planet and KE is its kinetic energy.

The total mechanical energy of the planet is **$KE + PE_g = 0.5 m V^2 - GmM / R$** **EQ1**

(m is the planet, M the sun, V is the speed of the planet . we suppose a circular orbit R)

But we also have the centripetal force = **$mV^2/R = GmM / R^2$** (gravitational force). **EQ2**

Play with EQ2 only and solve for $KE = 0.5 m V^2 = ?$ (as a function of PE_g . In magnitude)

☐

$PE_g/2$

☐

$PE_g/4$

☐

PE_g

☐

$2PE$

☒

Question 6 10 pts

You are pushing a mass of 10kg along a rough inclined plane. The angle is 30 degrees. the coefficient of friction is 0.5. The applied force (push) is 100N. The distance covered is 10m.

What is the net work done on the mass?

hint: the work done by the applied force is positive. the work done by gravity is negative . the work done by friction is negative.

the work done by gravity is $mgsin(30) \times \text{distance}$ or mgh same thing. and the normal force is $mgcos(30)$

This net work = change in KE. remember for test3

☐

about 47J

☐

about 17J

☐

about 27J

☐

about 67 J

Quiz saved at 2:36pm

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